

LeakBench-Tab: Separating Contamination Validity, Statistical Detectability, and Model Exploitability in Tabular Learning

Anonymous submission

Abstract

Prediction-time feature contamination can invalidate tabular model evaluation, yet existing evaluations often conflate three questions: whether a feature is invalid under the declared prediction boundary, whether a statistical diagnostic can locate it, and whether a model exploits it to inflate measured performance. We introduce a three-axis measurement framework that separates contamination validity (C), statistical detectability (D), and model exploitability (X). LeakBench-Tab instantiates this framework with 11 contamination mechanisms, five strength levels, eight model families, and 14,115 verified experiment cells organized into a 10,083-cell Core Tier and a 4,032-cell metadata-complete Meta Tier. Under the evaluated protocols, simple global contamination is readily localized and consistently exploited, whereas structured contamination is difficult to localize and exhibits low or model-dependent exploitability. Operational metadata improves in-domain localization within the Meta Tier but does not improve zero-shot transfer to the evaluated natural tasks; fixed field budgets also fail on redundant contamination clusters, while group- and lifecycle-based governance are not consistently superior to field-level removal. These findings show that invalidity, detectability, and exploitability are distinct measurement targets and should not be treated as a single leakage severity score.

1 Introduction

Prediction-time leakage occurs when a model is trained or evaluated with information that would not be valid at the declared prediction boundary. Direct copies of the target are obvious examples, but practical contamination can also arise through post-outcome aggregates, future-window statistics, or entity and source identifiers whose semantics depend on the deployment protocol. Such fields can make offline performance optimistic and undermine comparisons between models.

The presence of strong statistical association is not itself evidence that a feature is invalid. Conversely, an invalid feature need not be easy to locate from samples alone, and weak performance inflation does not make that feature valid. Treating these properties as interchangeable obscures both diagnostic failures and model behavior. We therefore

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study prediction-time contamination along three separate axes: contamination validity (C), statistical detectability (D), and model exploitability (X).

This paper asks three research questions. **RQ1:** Which forms of contamination can statistical diagnostics locate? **RQ2:** Which contamination mechanisms are exploited by different model families? **RQ3:** How do metadata and governance interventions affect localization and mitigation?

LeakBench-Tab addresses these questions through a controlled Core Tier, a metadata-complete Meta Tier, paired strict and permissive evaluation protocols, cross-model analysis, and boundary tests on natural tabular tasks. The benchmark is designed for measurement rather than as a claim of a universally superior detector or governance algorithm.

Our contributions are fourfold. First, we formalize C/D/X as separate measurement targets for prediction-time contamination. Second, we instantiate an 11-mechanism benchmark spanning five strengths and heterogeneous model coverage across Core and Meta tiers. Third, we identify a simple-structured divide and show that the apparent detectability-exploitability association is largely category-driven. Fourth, we show that operational metadata improves in-domain localization but not zero-shot natural-task transfer, and that structural governance is not uniformly superior to field-level auditing.

2 Related Work

3 Problem Setup and the C/D/X Framework

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ denote a supervised tabular prediction task with declared prediction time t_p . Feature j has a generation or availability time t_j and may additionally be invalidated by the declared data lineage or population boundary. The *strict* protocol retains only valid prediction-time features; the *permissive* protocol retains the injected or audit-confirmed contaminated fields.

3.1 Contamination Validity (C)

For feature j , we define

$$C_j = \mathbf{1}[j \text{ is invalid under the declared boundary at } t_p]. \quad (1)$$

Validity follows from the task boundary and mechanism construction. It is not determined by mutual information, feature attribution, model importance, or predictive performance.

3.2 Statistical Detectability (D)

Detectability is a feature-localization problem. A diagnostic produces a score s_j for each field, which is evaluated against C_j using feature-level area under the precision–recall curve (AUPRC), normalized AUPRC, and top- k recall. We use AUPRC as the main controlled-benchmark summary and rank-based measures when natural tasks contain very few contaminated fields.

3.3 Model Exploitability (X)

For model family m and metric q for which larger values are better, aligned harm is

$$X_m = q_m(\mathcal{D}_{\text{permissive}}) - q_m(\mathcal{D}_{\text{strict}}). \quad (2)$$

A positive value indicates measured performance inflation; a near-zero value indicates no material benefit under the evaluated protocol; and a negative value means the permissive protocol performs worse. A negative value is not evidence that the model actively rejects contamination.

4 LeakBench-Tab Design

4.1 Core Tier

The Core Tier evaluates 11 contamination mechanisms at five frozen strength levels. Table 1 groups them by the structure of their invalidity rather than by observed model performance.

4.2 Metadata-Complete Meta Tier

The Meta Tier supplies availability, structural, and environment context for controlled comparisons against statistical impact alone. We distinguish oracle metadata, operational raw metadata, operational derived metadata, and policy-equivalent metadata. Only raw and derived operational variants support claims about deployable metadata; policy-equivalent metadata is treated as an upper-bound condition.

4.3 Models and Natural-Task Boundary Tests

The five core families are logistic regression, random forest, CatBoost, LightGBM, and TabM. ModernNCA and TabR provide supporting audits, while TabPFN v2 remains exploratory because of limited coverage. Natural-task evidence comprises two evaluable boundary tests (Bank Marketing PRE and NYC 311) and one adapter-limited Lending Club entry that is excluded from transfer claims.

5 Experimental Protocol

The Core Tier contains 10,083 completed cells and the Meta Tier contains 4,032 cells; these tiers are not a uniform Cartesian product. Core comparisons use three seeds, paired strict/permissive evaluation, dataset-level evaluation, model-specific reporting, and equal-weight aggregation across the five core families. Statistical reporting is planned around paired effects, hierarchical bootstrap confidence intervals, and Holm correction within declared comparison families. These inferential outputs are not reported until they can be reconstructed from frozen source tables. For natural tasks with one contaminated field, we prioritize feature rank, top- k hits, and reciprocal rank over standalone AUPRC.

6 Results

- 6.1 Simple and Structured Mechanisms Occupy Different C/D/X Profiles
- 6.2 The Detectability–Exploitability Association Is Category-Driven
- 6.3 Exploitability Differs Across Model Families
- 6.4 Structural Governance Is Not Uniformly Superior
- 6.5 Operational Metadata Helps In-Domain but Does Not Transfer Zero-Shot

7 Limitations and Conclusion

LeakBench-Tab provides controlled ground truth but relies heavily on semi-synthetic tasks. Natural-task contamination labels are sparse and single-reviewer, the Lending Club entry is adapter-limited, model coverage is intentionally tiered, and the current evidence does not establish a causal relationship between model capacity and exploitability. Within these limits, the benchmark shows that prediction-time invalidity, statistical detectability, and model exploitability answer different questions. Operational context can improve localization in-domain without transferring zero-shot, and more structured governance does not automatically provide better mitigation. Evaluations of tabular contamination should therefore report C, D, and X separately and scope conclusions to the protocols under which each is measured.

ID	Mechanism	Category	Invalidity source / structure
M01	Direct target copy	Simple	Target-derived global field
M02	Noisy target proxy	Simple	Noisy target-derived global field
M06	Redundant leakage cluster	Simple	Multiple correlated target proxies
M10	Mixed leakage	Simple	Legitimate and target-derived components
M03	Nonlinear target transform	Boundary	Nonlinear target-derived field
M07	Sparse subgroup leakage	Boundary	Contamination restricted to a subgroup
M11	Graph-mediated leakage	Boundary	Correlated contamination component
M04	Post-outcome aggregation	Structured	Post-outcome temporal aggregate
M05	Temporal look-ahead	Structured	Future-window information
M08	Entity leakage	Structured	Entity-conditioned information
M09	Source leakage	Structured	Source-conditioned information

Table 1: LeakBench-Tab mechanism taxonomy. Detailed parameterizations and strength mappings are reserved for the supplement.